Project Management Plan

Upstream Scenario Modelling Framework

Finance

Project Definition, Objectives, and Scope

Objective: Develop and evaluate a controlled, transparent and reproducible framework for constructing and analysing alternative Upstream portfolio scenarios.

Business Need

Portfolio scenario analysis relied heavily on repeated manual spreadsheet manipulation, making alternative cases time-consuming to construct and making reproducibility, transparency and interrogation more difficult.

Project Response

Design and implement a controlled scenario modelling framework combining governed enterprise and manual planning data, an Excel user interface, and a modular Python analytical engine. Users can select modelling cases and apply reusable scenario operations while retaining traceability to the underlying planning data.

Intended Business Value

Improve optionality, transparency, repeatability and the ability to interrogate alternative portfolio scenarios, while preserving the role of expert business judgement.

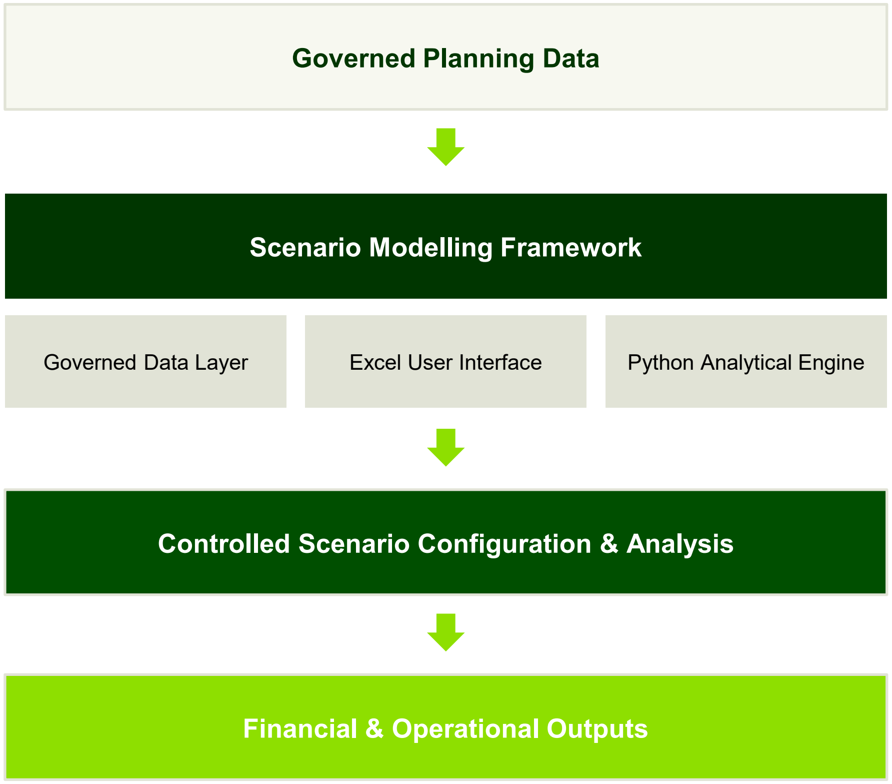
In Scope

- Governed integration of planning data
- Financial Entity to modelling-case translation
- Controlled manual planning-data integration
- Scenario configuration and reusable transformation operations
- Financial and operational scenario outputs
- Validation, reporting and user controls

Deliberately Outside Current Scope

- Predictive or machine-learning forecasting
- Fully industrialised enterprise deployment
- Commodity-price functionality without a validated business methodology
- Fully validated country-specific fiscal/tax modelling
- Portfolio optimisation as an implemented production capability

Planned solution concept



Indicative solution boundary as defined at project outset; not an implemented system.

Project Delivery Timeline and Workstreams

Structured delivery across business support, academic outputs, and parallel application development

Phased delivery — Defined phases from discovery and design through build, validation and final evaluation.

Parallel workstreams — Business support, DT603 academic delivery and Streamlit development progress in parallel.

Milestone-led control — Progression governed by validation gates and formal submission checkpoints.

	1 Discovery and re-engagement Weeks 1–2	2 Terms of Reference and scope Week 3	3 Baseline data and mapping Weeks 3–4	4 Scenario build and configuration Weeks 4–6	5 Pilot evaluation and SME validation Weeks 6–8	6 Final evaluation and recommendation Weeks 8–10
Business / LTP process support	• Review earlier prototype work • Re-engage Upstream portfolio stakeholders • Confirm business need	• Refresh terms of reference • Agree SME user group • Prioritise scenario operations	• Prepare trusted baseline dataset • Agree approach to financial entity aggregation	• Build, test, and validate priority scenario operations	• Run alongside live LTP business process • Reconcile to BP referenced views	• Assess usability, traceability, scalability • Document limitations • Recommend future development path
DT603 academic delivery	• Confirm DT603 scope, outcomes, and delivery approach	• Create project management plan • Set-up repository • Set-up meeting cadence	• Capture extraction, profiling, and architecture evidence	• Develop core working data analysis solution	• Capture business validation evidence • Conduct stakeholder walkthroughs • Gather feedback	• Complete DT603 submission package
Parallel Streamlit app development	• Environment set-up and tooling review • Gather app specific requirements	• Set-up VS code workspace • Learn agentic development approach	• Build Streamlit scaffold, navigation, and scenario input forms	• Connect the Streamlit app to the Python scenario engine • Add charts and comparison views	• Demo the app • Refine UI/UX • Capture feedback	• Recommend next phase development path
Milestones	**Context & feasibility confirmed**	**Terms of Reference approved**	**Baseline & mapping approach agreed**	**Priority scenarios configured**	**Validation evidence captured**	**Final submission and recommendation**

Integrated delivery timeline showing project phases, workstreams, milestones, and expected outputs

Governance, Stakeholders and Delivery Control

Planned governance structure, delivery accountabilities and structured decomposition of work

Project Governance

Ale
Project Sponsor — senior business direction

Pam
Senior stakeholder — business oversight

Dan Waters
Line Manager — delivery governance

Risk escalation route

Dan Deaton
Project Lead — solution design and delivery

Risk escalation route

Galina
Business SME

Cathy
Business SME

Deepak
Data SME

Archana
Data SME

Subject Matter Experts — Galina, Calvin, Deepak, and Archana provide requirements clarification, data and methodology expertise, validation and challenge.

Structured Delivery Control

EPIC Deliver Upstream Scenario Modelling Framework

Governed Data Foundation
Enterprise and manual data integration, financial measures, modelling-case translation

Scenario Configuration & User Controls
Case selection, scenario parameters, validation and application controls

Analytical Engine
Reusable Python scenario operations, financial calculations and controlled outputs

Testing & Business Validation
Functional testing, reconciliation, SME review and performance testing

Documentation & Delivery Evidence
Architecture, repository, user guidance, validation evidence, DT603 artefacts

User Stories → Tasks decomposed into testable delivery increments with defined ownership and evidence.

Governance and Communication Controls

Sponsor governance
direction, success criteria and key decisions

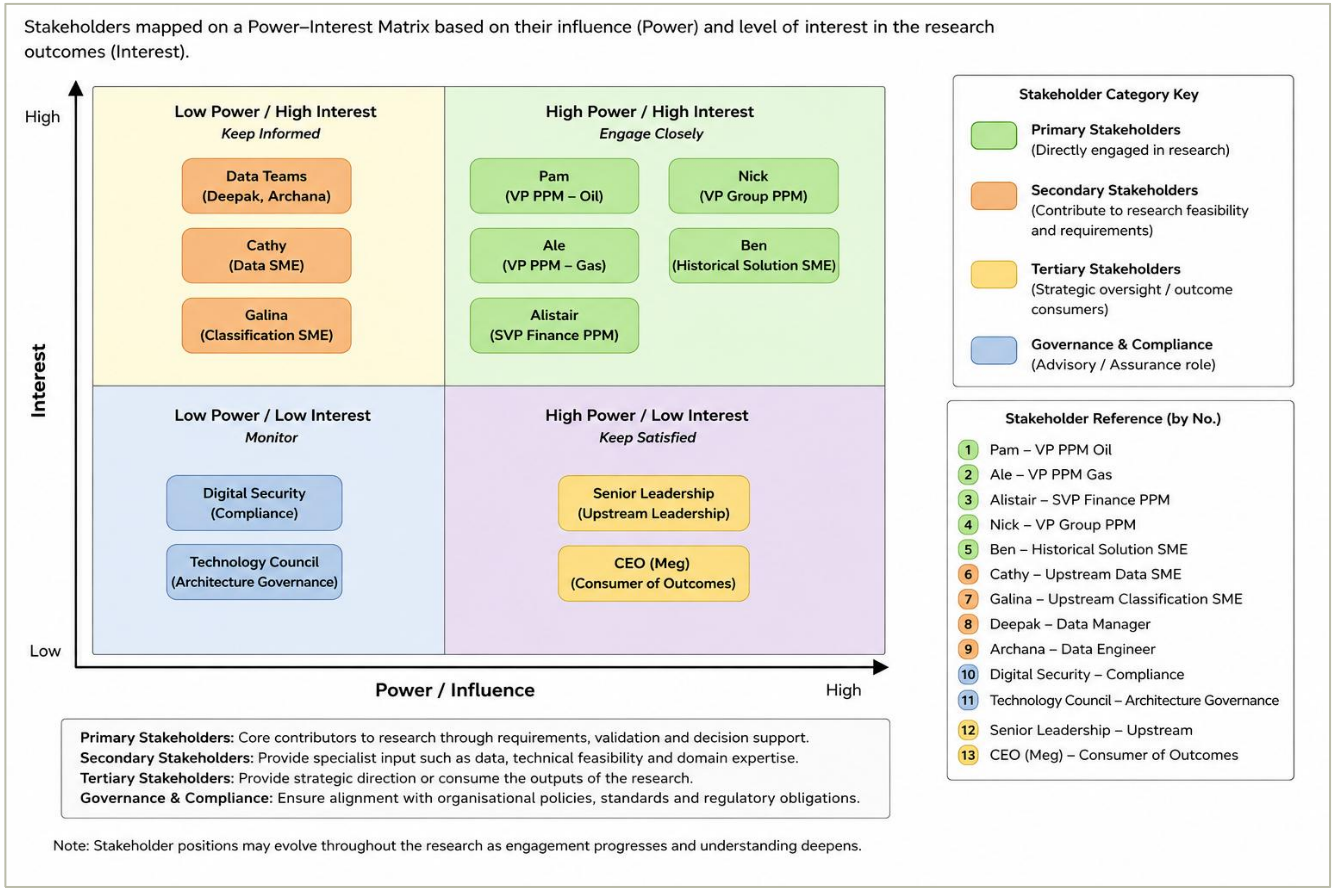
SME engagement
requirements, business rules, data understanding, validation

Delivery tracking
prioritised work decomposition and progress control

Communication & assurance
planned reviews, walkthroughs and evidence capture

Stakeholder Engagement and Communication Strategy

Mapping stakeholder influence and interest to plan proportionate engagement



Stakeholder power–interest matrix – stakeholders positioned by influence over, and interest in, the project.

Engagement intensity is planned according to stakeholder influence, interest and contribution.

Stakeholder Mapping and Analysis

- Assess stakeholders by relative power and interest
- Define the appropriate level of engagement
- Focus attention on those with the greatest influence on project direction, requirements, validation, and delivery

Engagement Classification

Primary Sponsor, senior business stakeholders and line management – managed closely

Secondary Subject matter experts and delivery contributors – kept satisfied and consulted

Tertiary Wider planning community and governance functions – kept informed

Critical Engagement

Sponsor and senior stakeholder involvement secures direction, prioritisation and access to governed data and expertise

Risk, Dependencies and Mitigation

Planned controls for the principal threats to successful project delivery

#	Threat	Gap	Category	L	I	Rating	Mitigation
1	**Misinterpretation of stakeholder requirements**	Capability	Technical	H	H	**H**	Structured operation definitions and stakeholder validation
2	**Fragmented planning and financial data**	Capability	Technical	H	H	**H**	Data profiling, reconciliation and standardised transformation rules
3	**Financial logic validation challenges**	Governance	Technical	M	H	**H**	SME review, testing and reconciliation against existing outputs
4	**Organisational change and shifting priorities**	Capability	Org	M	H	**H**	Iterative requirements review and stakeholder engagement
5	**Competing initiatives (System of Performance, Horizon X, Lighthouse)**	Capability	Org	M	M	**M**	Scope management and sponsor alignment
6	**Mishandling commercially sensitive planning information**	Governance	Ethical / Legal	L	H	**M**	bp governance controls and data classification policies
7	**Researcher bias arising from insider position**	Governance	Research	L	M	**M**	Triangulation and stakeholder validation
8	**Dependency on specialist knowledge**	Sustainability	Org	M	H	**H**	Knowledge transfer, documentation and transparent logic
9	**Framework perceived as overly complex**	Sustainability	Adoption	M	M	**M**	User-centred design, training, documentation
10	**Prototype-led development evolves into enterprise platform expectations**	Capability	Delivery	M	H	**H**	Maintain clear scope boundaries and governance process for enhancements

Project risk log – threats assessed by likelihood and impact, with planned mitigation. Ratings shown are as assessed at planning stage.

Risk Management Approach

Identify Threats captured across capability, governance, organisational and delivery dimensions.

Assess Each risk scored for likelihood and impact on a 5×5 matrix to set priority.

Mitigate Proportionate controls agreed and built into the delivery plan and ways of working.

Monitor Risks reviewed through project governance, with escalation to the sponsor where required.

Key Project Dependencies

- Timely access to governed planning and financial data
- Subject matter expert availability for review and validation
- Continued sponsor and line management support
- Approved bp environments for sensitive planning data
- Scope stability alongside competing initiatives

Risk control is integrated into delivery planning, with material risks assessed, mitigated and monitored through project governance.

Success Criteria, Validation and Project Closure

The project will be evaluated across the following dimensions to assess both the technical performance of the prototype and the value it delivers to the wider planning process.

Evaluation Dimension	Objective	Measure / Evidence
1 **Technical Capability**	Demonstrate that the prototype reliably performs the intended scenario modelling functions.	Successful execution of scenario operations, validated data pipeline, repeatable outputs and reconciliation to trusted planning artefacts. Ability to configure and compare multiple portfolio scenarios, including manual inputs, portfolio construction and optimisation capability.
2 **Business Value**	Improve planning efficiency, repeatability and decision support.	Estimated reduction in manual effort, increased consistency of scenario construction and SME assessment of planning usefulness.
3 **Governance & Transparency**	Demonstrate trusted and auditable analytical processes.	Documented assumptions, mappings, transformation rules, validation evidence, controlled manual inputs and security considerations.
4 **Adoption & Usability**	Confirm that planners can understand, operate and challenge the prototype.	Walkthrough feedback, usability observations, review comments and enhancement requests.
5 **Strategic Contribution**	Assess the prototype's contribution beyond the immediate use case.	Evidence of reusable architecture, lessons learned and contribution to wider initiatives (e.g. Horizon X).
6 **Sustainability**	Assess whether the approach can be maintained, extended or scaled.	Recommendations for future development, known limitations and suitability for tactical or enterprise implementation